Exercitiu $f : U \to V$ izomorfism

$u_1 \dots u_n \in U$

$\Rightarrow$

a) $u_1 \dots u_n$ liniar independenti in $U \iff f(u_1) \dots f(u_n)$ liniar independenti in V

b) $u_1 \dots u_n$ sistem de generatori pt $U \iff f(u_1) \dots f(u_n)$ sistem de generatori pt. V

c) $u_1 \dots u_n$ baza in $U \iff f(u_1) \dots f(u_n)$ baza in V.

Einstein

Exemplu $f : \mathbb{R}_2[X] \to M_2(\mathbb{R})$

$$f(P) = \begin{pmatrix} P(1) & P'(1) \\ 0 & P(1) + P'(1) \end{pmatrix}$$

$\mathcal{B} = \{1, X, X^2\}$

$\mathbb{R}_2[X]$ $\dim_{\mathbb{R}} \mathbb{R}_2[X] = 3$

$B' = \{E_{11}, E_{12}, E_{21}, E_{22}\}$

$M_2(\mathbb{R})$ $\quad \dim M_2(\mathbb{R}) = 4$

$$M(f)_{B'}^{B} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 1 & 2 & 3 \end{pmatrix} \in M_{4,3}(\mathbb{R})$$

$$f(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \cdot E_{11} + 0 \cdot E_{12} + 0 \cdot E_{21} + 1 \cdot E_{22}$$

$$f(x) = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} = 1 \cdot E_{11} + 1 \cdot E_{12} + 0 \cdot E_{21} + 2 \cdot E_{22}$$

$$f(x^2) = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = 1 \cdot E_{11} + 2 \cdot E_{12} + 0 \cdot E_{21} + 3 \cdot E_{22}$$

$$\begin{array}{ccc} R_2[x] & \xrightarrow{f} & M_2(\mathbb{R}) \\ P(x) = a + bx + cx^2 & & f(P) = \begin{pmatrix} a+b+c & b+2c \\ 0 & a+2b+3c \end{pmatrix} \\ \downarrow & & \downarrow \\ \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 1 & 2 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3 & \xrightarrow{M(f)} & \begin{pmatrix} a+b+c \\ b+2c \\ 0 \\ a+2b+3c \end{pmatrix} \in \mathbb{R}^4 \checkmark \end{array}$$

$$U \xrightarrow{f} V \xrightarrow{g} W$$
$$n \qquad\quad m \qquad\quad p$$

$M(f)$ (m, n) $\qquad$ $M(g)$ (p, m)

$$M(g) \cdot M(f) = M(g \circ f)$$
$$(p, m)\ (m, n) \qquad (p, n)$$

U: $B = \{u_1, \dots u_n\}$ $\qquad M(f) = (a_{ji})_{\substack{j=\overline{1,m} \\ i=\overline{1,n}}}$

V: $B' = \{v_1, \dots v_m\}$

W: $B'' = \{w_1, \dots w_p\}$ $\qquad M(g) = (b_{kj})_{\substack{k=\overline{1,p} \\ j=\overline{1,m}}}$

$$f(u_i) = \sum_{j=1}^{m} a_{ji} v_j, \quad i = \overline{1,n}$$

$$g(v_j) = \sum_{k=1}^{p} b_{kj} w_k, \quad j = \overline{1,m}$$

$M(g \circ f)$

$g \circ f(u_i) = g(f(u_i)) = g\left(\sum_{j=1}^{m} a_{ji} v_j\right) =$ $\quad i = \overline{1,n}$

$= \sum_{j=1}^{m} a_{ji} g(v_j) = \sum_{j=1}^{m} a_{ji} \sum_{k=1}^{p} b_{kj} w_k =$

$= \sum_{k=1}^{p} \left(\sum_{j=1}^{m} b_{kj} a_{ji} \right) w_k \longrightarrow (M(g)M(f))_{ki}$

$M(g \circ f)_{ki}$

$\Rightarrow M(g \circ f) = M(g) M(f)$

$f : U \to V$, aplic. liu. $n = \dim_{\mathbb{K}} U = \dim_{\mathbb{K}} V$

f izomorfism $\iff$ $M(f)$ inversabilă.

$\Rightarrow$ " f izom $\Rightarrow \exists f^{-1}$ aplic. liu.

$f \circ f^{-1} = id_V$ $\quad V \xrightarrow{f^{-1}} U \xrightarrow{f} V$

$f^{-1} \circ f = id_U$

$M(f)\,M(f^{-1}) = M(id_V) = I_n$

$M(f^{-1})\,M(f) = M(id_U) = I_n$

"$\Leftarrow$" f izomorfism $\Leftarrow$ inj. $\ker(f) = \{0\}$

surj. $\operatorname{Im}(f) = V$

$\underbrace{\dim \ker(f)}_{0} + \dim \operatorname{Im}(f) = n$

$\Rightarrow$ f surjectivă $\Rightarrow$ f izom.

Fie $u \in \ker(f) \Rightarrow f(u) = 0_V$

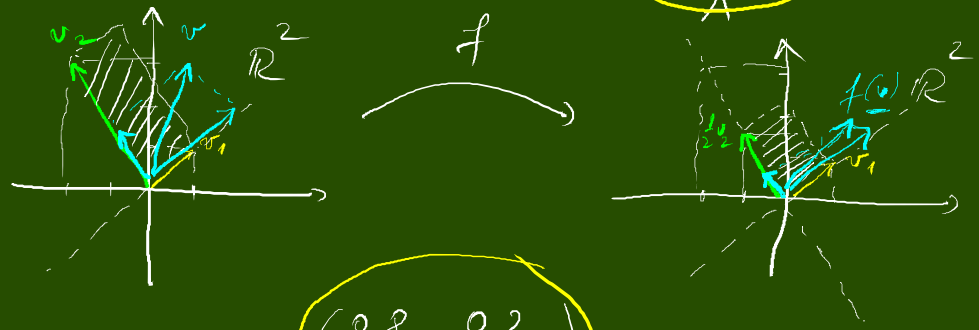
$\Rightarrow \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = [f(u)]_{B'} = M(f)_{B'}^{B}\,[u]_B \quad / \cdot M(f)^{-1}$

$\Rightarrow \quad [u]_B = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad \Rightarrow \quad u = 0_U$

$\Rightarrow \quad \ker(f) = \{0_U\} \quad \Rightarrow \quad f$ inj.

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}^2$ aplicație liniară

$$f\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0{,}8x + 0{,}2y \\ 0{,}3x + 0{,}7y \end{pmatrix} = \underbrace{\begin{pmatrix} 0{,}8 & 0{,}2 \\ 0{,}3 & 0{,}7 \end{pmatrix}}_{A} \begin{pmatrix} x \\ y \end{pmatrix}$$



$$M(f)_B = \begin{pmatrix} 0{,}8 & 0{,}2 \\ 0{,}3 & 0{,}7 \end{pmatrix} = A$$

$B = \{e_1, e_2\}$ bază canonică

$f(e_1) = \begin{pmatrix} 0{,}8 \\ 0{,}3 \end{pmatrix} = 0{,}8e_1 + 0{,}3e_2$

$f(e_2) = \begin{pmatrix} 0{,}2 \\ 0{,}7 \end{pmatrix} = 0{,}2e_1 + 0{,}7e_2$

$B' = \{ v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \}$

bază ?

$M(f)_{B'} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ matrice diagonală.

$f(v_1) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = v_1 = 1 \cdot v_1 + 0 \cdot v_2$

$f(v_2) = \begin{pmatrix} -1 \\ 1,5 \end{pmatrix} = \frac{1}{2} v_2 = 0 \cdot v_1 + \frac{1}{2} \cdot v_2$

$f(v_1) = Av_1 = v_1$ $\qquad$ $f(v_2) = Av_2 = \frac{1}{2} v_2$

eigenvalue

eigenvector

$f(v) = Av = \lambda v$

$V \xrightarrow{f} V$ $\qquad$ 2 baze B, B'

$M(f)_B = ?$ $\qquad$ $M(f)_{B'} ?$

$V \xrightarrow[B']{} V \xrightarrow[B]{id_V} V \xrightarrow[B]{f} V$ $\qquad$ $f \circ id_V = f$

$V \xrightarrow[B']{f} V \xrightarrow[B]{id_V} V$ $\qquad$ $id_V \circ f = f$

$$f \circ id_V = id_V \circ f$$

$$M(f \circ id_V)_{B'}^{B} = M(id_V \circ f)_{B'}^{B}$$

$$M(f)_B^B \underbrace{M(id_V)_B^{B'}}_{P_B^{B'}} = \underbrace{M(id_V)_B^{B'}}_{P_B^{B'}} M(f)_{B'}^{B'}$$

$$M(f)_B \cdot P = P \cdot M(f)_{B'}$$

(diagonalizable) (diagonal)

$$\underbrace{M(f)_B}_{A} = P \cdot \underbrace{M(f)_{B'}}_{D} \cdot P^{-1}$$

$$A^n = A \cdots A = P D P^{-1} P D P^{-1} \cdots P D P^{-1}$$

$$= P \cdot D^n \cdot P^{-1}$$

$$A = \begin{pmatrix} 0{,}8 & 0{,}2 \\ 0{,}3 & 0{,}7 \end{pmatrix}^n = P \begin{pmatrix} 1^n & 0 \\ 0 & \frac{1}{2^n} \end{pmatrix} P^{-1}$$

$$P = P_B^{B'} = (v_1 \mid v_2) = \begin{pmatrix} 1 & -2 \\ 1 & 3 \end{pmatrix}$$

tema: verificare $A = P \cdot D \cdot P^{-1}$

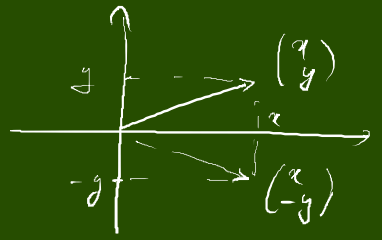
$(AP = PD)$

$A^n = P \cdot D^n \cdot P^{-1}$

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2$$

(AB e BA: $2AB$)

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



$$A\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ -y \end{pmatrix}$$

$\begin{pmatrix} x \\ y \end{pmatrix}, \begin{pmatrix} x \\ -y \end{pmatrix}$ coliniari $\Leftrightarrow$?

$y = 0$ $\quad A\begin{pmatrix} x \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix}$

$\lambda_1 = 1$

$x = 0$ $\quad A\begin{pmatrix} 0 \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ -y \end{pmatrix}$

$\lambda_2 = -1$

$\sigma(A)$ spectrul matricei A

$\lambda \in \sigma(A) \longrightarrow V_\lambda = \{ v \in K^n \mid Av = \lambda v \}$

$\sigma(A) = ?$ $\quad V_\lambda = ?, \lambda \in \sigma(A)$

λ val pr. $\Leftrightarrow$ $\exists\, v \neq 0$ a.i. $Av = \lambda v$

$Av - \lambda v = 0$

$(A - \lambda I_n) v = 0 = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ Sistem cu n ec. cu n necun. omogen

Sistemul are sol. nebanale $\Leftrightarrow$ $\det(A - \lambda I_n) = 0$

$P_A(X) = \det(A - XI_n)$

$n = 2$ $\quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$P_A(X) = \begin{vmatrix} a - X & b \\ c & d - X \end{vmatrix} =$

$$= (a - X)(d - X) - bc$$

$$= X^2 - \underbrace{(a+d)}_{tr(A)} X + \underbrace{(ad - bc)}_{\det(A)}$$

n arbitrar

$P_A(X) \in k[X]$ $\quad P_A(X) = (-1)^n X^n + \dots + \det(A)$

grad $P_A = n$

coeficientul lui X^n este $(-1)^n$

Rădăcinile polin. caract. sunt exact val. pr. $\lambda_1 \dots \lambda_n$ $\quad \sigma(A)$

$\xrightarrow{\text{rel. Viète}}$ $\lambda_1 + \dots + \lambda_n = tr(A)$

$\dots\dots\dots$

$\lambda_1 \dots \lambda_n = \det(A)$

Thm (Hamilton - Cayley) $\quad A \in M_2(k)$

$A^2 - tr(A)A + \det(A) I_2 = O_2$

$A \in M_n(\mathbb{R}) \Rightarrow P_A(A) = O_n$

$$\boxed{P_A(X) = \det(A - XI_n)}$$

Aplicații: A^n, A^{-1}

$$\Rightarrow 1 \cdot A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I_n = O_n$$

$A^n, A^{n-1}, A^{n-2}, \dots, A, I_n$ liniar dep.

$$A^n = -a_1 A^{n-1} - a_2 A^{n-2} \dots - a_n I_n$$

$$A^{n+1} = -a_1 A^n - a_2 A^{n-1} \dots - a_n I_n$$

$$\in Sp\{A^{n-1}, \dots, A, I_n\}$$

$\forall k \in \mathbb{Z}$ $\quad A^k \in Sp\{A^{n-1}, \dots, A, I_n\}$

$\exists A^{-1} \Rightarrow \quad A^n + a_1 A^{n-1} + \dots + a_n I_n = O_n$

$\neq 0$

$-a_n / \quad -a_n I_n = A\left(A^{n-1} + a_1 \cdot A^{n-2} + \ldots + a_{n-1} I_n\right)$

$$I_n = A \cdot \underbrace{\left(\qquad\qquad\qquad